Sahaj Bhadja

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Summary

Software engineer with 2+ years building **AI-enabled internal tools** in **Python/Java**, integrating **GenAI models** into production workflows with **microservices**, **SQL** schemas, and **DevOps** CI/CD. Experienced in **Agile** delivery, automated testing, monitoring, and cross-functional stakeholder collaboration for transformation-focused engineering teams.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Software Engineering, Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Object-Oriented Programming, Software Engineering, Data Structures, Operating Systems

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer - Transformation *September 2025 – Present*

- Designed **AI-first Python/FastAPI** applications and orchestration tools for 500+ internal users, integrating **GenAI models** into async pipelines processing 5K+ monthly jobs with **PostgreSQL/SQL Server** schemas and seamless cross-system data flows.
- Built automated validation and monitoring for production releases, leveraging **GenAI-assisted** documentation and test workflows to sustain **99%+ reliability** while partnering with stakeholders on requirements, progress updates, and process-improvement recommendations.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **GenAI-enabled** grading tools using **GPT-4o, Claude, and Qwen** models in an **Agile** startup, integrating multimodal AI into **Python/Flask MVC** services handling **10K+ daily requests** for internal instructor stakeholders.
- Reduced workflow turnaround by **60%** through AI-assisted automation, unit-test coverage, and iterative delivery with product managers while using coding agents to improve maintainability, documentation quality, and engineering productivity.

USC AD Labs **Los Angeles, CA**
Software Engineering Assistant *January 2025 – May 2025*

- Implemented internal data-processing tools across 100K+ records with reproducible automated tests, technical documentation, and transparent communication supporting cross-functional research teams adopting iterative, outcome-driven development practices.
- Identified workflow automation opportunities and contributed enhancement recommendations through structured reviews, defect analysis, and collaborative planning with senior engineers across complex multi-application research environments.

Aasma Technology Solutions **India**
Software Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **Java/Python microservices** with **MVC**-style service boundaries and AWS integrations, improving API response by **30%** while supporting transparent data movement, metrics-friendly reporting, and maintainable internal platform delivery patterns.
- Maintained **20+ automated test suites** in **Jenkins DevOps** pipelines, increasing release confidence by **68%** through unit/integration automation, monitoring-ready logs, and continuous improvement of repetitive manual engineering workflows.

Aasma Technology Solutions **India**
Software Engineer Intern - (VibrantChem) *May 2023 – August 2023*

- Built **microservice** dashboard modules with relational **SQL** schemas and real-time updates across 2M+ records, improving operational visibility and decision-support metrics for business-unit stakeholders through responsive internal web tooling.
- Automated **Docker/Jenkins** deployments, cutting release cycles by **50%** while collaborating with developers and QA partners on documentation, user-support readiness, and Agile-style iterative enhancements to internal customer-facing applications.

Skills

Languages & Architecture: **Python**, **Java**, C++, Ruby (familiar), OOP, **MVC**, **microservices**, REST APIs, FastAPI, Flask, Spring Boot exposure
AI & Automation: **GenAI**, **LLM integration**, GPT-4o, Claude, OpenAI APIs, RAG/embeddings, AI agents, automated unit testing, Cursor, GitHub Copilot, process automation
DevOps & Data: **Agile/Scrum**, **DevOps**, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, PostgreSQL, SQL Server, MySQL, SQL, monitoring, documentation, stakeholder communication, metrics-driven delivery, Git, AWS, Azure

Projects

Big-Brain Live | *GenAI, LLaMA-2, FAISS, Internal Tools*

- Built an **AI-first** document platform integrating **LLaMA-2** models with semantic search over 50K+ records, reducing analysis time by **60%** through GenAI summarization, monitoring-friendly workflows, and collaborative internal-user experiences.

Linguistix Code | *Python, AWS, Docker, Pipeline Automation*

- Deployed automated **Python** data-pipeline services on **AWS** with **Docker**, validation gates, and CI-style releases, streamlining repetitive backend tasks and improving throughput for transformation-style internal processing and integration workflows.

CaptionCraft | *Python, GenAI, Model Integration, Monitoring*

- Integrated generative AI captioning models into a production-style application with structured error handling, testable service layers, and user-experience monitoring patterns supporting reliable model adaptation in software delivery environments.